EdgeMonitor Gateway Architecture and API Notes

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1. Architecture Overview

EdgeMonitor Gateway is composed of local services running on one industrial gateway.

Main services:

- Device Manager;
- Protocol Adapter Runtime;
- Telemetry Normalizer;
- Local Buffer;
- Cloud Forwarder;
- Audit Service;
- Diagnostic Service.

The gateway shall continue collecting telemetry even when the cloud ingestion endpoint is unavailable.

2. Device Manager

The Device Manager stores registered devices and validates device configuration.

Device configuration fields:

- device_id;
- display_name;
- protocol;
- host;
- port;
- polling_interval_seconds;
- signal_map;
- enabled;
- configuration_version.

The Device Manager publishes a configuration change event after each successful update.

3. Protocol Adapter Runtime

The Protocol Adapter Runtime starts one adapter per enabled device.

Supported adapters for the pilot:

- ModbusTcpAdapter;
- MqttSubscriberAdapter;
- HttpJsonSensorAdapter.

If an adapter crashes three times within five minutes, the gateway shall mark the device as degraded and generate an operational alert.

4. Telemetry Normalizer

The Telemetry Normalizer receives raw protocol payloads and emits normalized telemetry records.

If a payload does not match the configured signal map, the record shall be rejected and a validation error shall be logged.

Rejected payloads shall not be forwarded to the cloud.

5. Local Buffer

The Local Buffer stores normalized telemetry before cloud forwarding.

Buffer behavior:

- records are ordered by original_timestamp then ingestion_timestamp;
- records are marked forwarded only after cloud acknowledgement;
- retry_count is tracked per record;
- records are not removed before successful forwarding unless retention limits are exceeded.

6. Cloud Forwarder

The Cloud Forwarder sends telemetry batches to the cloud ingestion endpoint.

Retry policy:

- initial retry delay: 1 second;
- multiplier: 2;
- maximum retry delay: 60 seconds;
- alert after 5 consecutive forwarding failures.

The forwarder shall use event_id as an idempotency key.

7. API Summary

POST /api/v1/devices

Creates a device.

Required role:

- maintenance_engineer;
- site_admin.

Expected responses:

- 201 Created;
- 400 Invalid configuration;
- 401 Unauthenticated;
- 403 Forbidden;
- 409 Duplicate device ID.

GET /api/v1/devices

Returns registered devices and health status.

Required role:

- plant_operator;
- maintenance_engineer;
- site_admin.

PATCH /api/v1/devices/{device_id}

Updates device configuration and creates a new configuration version.

POST /api/v1/configuration/rollback

Rolls back to the previous valid configuration version.

Required role:

- site_admin.

POST /api/v1/diagnostics/export

Creates a diagnostic export.

Required role:

- site_admin.

Sensitive values shall be redacted before the file is written.

8. Observability

The gateway exposes:

- `/health/live`;
- `/health/ready`;
- adapter status;
- buffer capacity metric;
- forwarding retry metric;
- audit event stream.

9. Architecture Risks

Known risks:

- duplicate telemetry after partial cloud timeout;
- delayed device health transition for MQTT disconnects;
- accidental leakage of credentials in diagnostic logs;
- misconfigured polling intervals causing excessive load.